



# ***Zero knowledge bridges***

A learning group for ZK and SNARK application development

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Respectful collaboration starts here

**ALL ARE  
WELCOME**

**TODO  
BIENVENIDOS**

**みなさん  
歓迎します。**

[www.lfdecentralizedtrust.org/code-of-conduct](http://www.lfdecentralizedtrust.org/code-of-conduct)

# Agenda



- Logistics
- Zero knowledge proofs - summary
- Interoperability basics: Cross-chain token
- Interoperability basics: Atomic cross-chain swap
- Blockchain bridge basics
- Blockchain bridge with cryptographic guarantee
- ZK bridge
- ZK bridge projects
- Challenge
- Links, Resources, Literature
- Q&A

# Logistics

Every month or two months

Similar structure, similar logistics + **guest lectures**

## Theory:

- Look tables,
- Zk variations: STARKs, FRI
- recursive SNARKs

## Programming / platforms:

- ZoKrates, Leo
- Zinc / ZKSync

## Applications:

- ZK bridges
- ZK Virtual Machines
- ZK Identity / verifiable credential - presentations



## Discord channel:

<https://discord.com/channels/905194001349627914/1329201532628898036>

## Meetup.com:

<https://www.meetup.com/lfdt-hungary/events/305634614/>

## Repo with all the contents:

<https://github.com/LF-Decentralized-Trust-labs/zk-learning-group>

# Zero knowledge proofs - summary

## Verifiable computation with cryptographic proof

### Scarce resource:

- limited verification time: no re-execution
- storage limits: smaller proof
- limited information

**Computation:** arithmetic circuit :  $C(x, w) \rightarrow F$

- $x$  public input
- $w$  private input, witness
- high level computation
- arithmetic circuit
- polynomials

**Prover** algorithm:  $P(pp, x, w) \rightarrow \text{proof}$

**Verifier** algorithm:  $V(vp, x, \text{proof}) \rightarrow \text{accept / reject}$

## Zero Knowledge Proof



Interactive / non-interactive proofs

# Interoperability basics: *Cross-chain Token*

Cross-chain information: same information on both chains

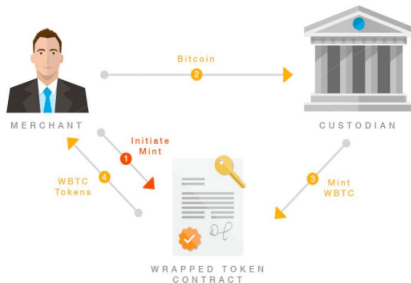
Cross-chain double spending: the token can be spent on the one or other chain, but not both

**Lock and mint:** token is locked on one chain and minted on the other one

**Burn and unlock:** the other token is burned on the other chain and the locked is unlocked on the first.

**Intermediaries:** listening the event on one chain and triggering mechanism on the other one:

- Centralized / decentralized



# Interoperability basics: *Atomic cross-chain swap*

A transaction happens on one chain if and only if a second one on another chain

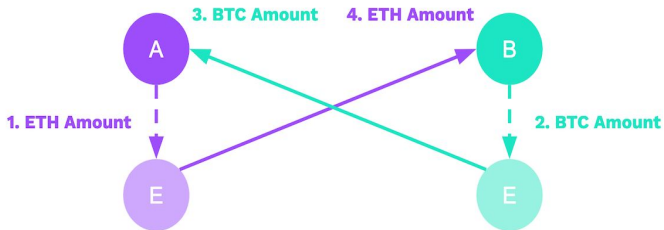
Executing two transfers simultaneously:  
either two or non will be executed

HTLC: Hashed timelock contract:

- Temporal transaction, unlocking conditions
- **Forward:** preimage of the hash function
- **Backward:** timelock expires

Cross-chain swap = 2 HTLC + intermediate components

Centralized and decentralized



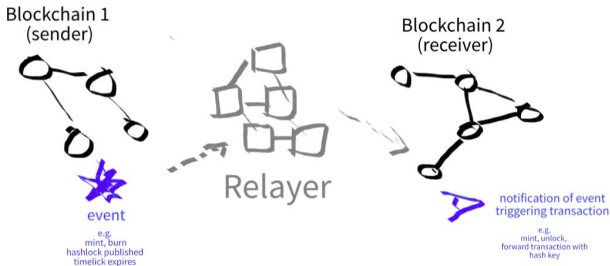
# Blockchain bridge basics

General interoperability problem: Sender and relayer chain: an event happens on the C1 **sender chain** and C2 **receiver chain** must be notified

**Relayer:** delegating the event from a sender chain to the receiver chain:

- Centralized relayer
- Decentralized relayer:
  - Economic guarantees (like Polynetwork, Wormhole, Ronin)
  - Cryptographic guarantees
  - Hybrid
  - Built-in bridges and relayers (like Polkadot)

Different security guarantees, different privacy properties



# Blockchain bridge with cryptographic guarantee

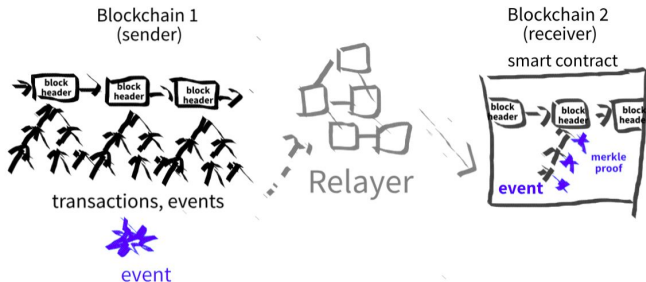
Simplified payment verification, light client

Maintaining block headers in a cryptographic consistent way, but without the transactions or events.

Proving if a transaction or event is in the blockchain: merkle inclusion proof

Receiver blockchain stores and validates block headers of the source chain and can validate events of the source chain with merkle inclusion proof.

Storing and validating block headers or merkle proofs in a smart contract is very resource intensive

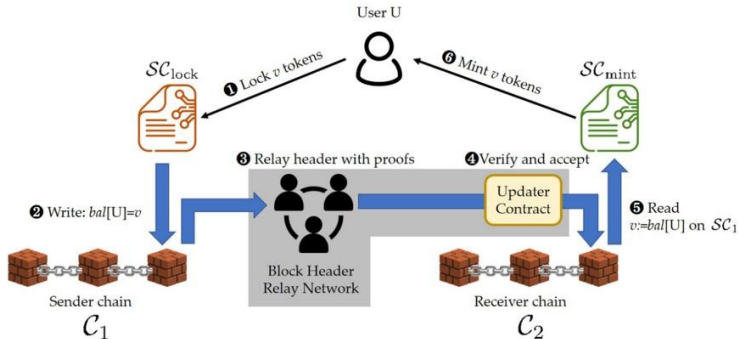


# Zero knowledge bridge

Cryptographic proofs  
for the new headers  
and merkle tree  
inclusion is generated  
off-chain by the relayer  
network: zero  
knowledge proof

Smart contract  
generated the  
correctness of ZK proof

Proof verification is  
much less resource  
intensive



# ZK bridge projects

## Succinct Labs:

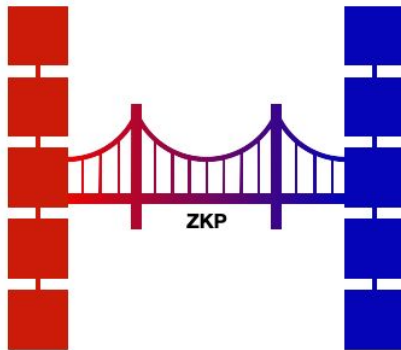
- ZK light client + consensus proof
- SP1 zkVM: a general-purpose, RISC-V-based zero-knowledge virtual machine
- General framework for building cross-chain application

## zkIBC by Electron Labs:

- ZK protocol that brings the Cosmos Inter-Blockchain Communication (IBC) protocol to the Ethereum blockchain

## zkBridge by Polyhedra Network:

- General cross-chain protocol based on ZKP
- Connecting 25-30 L1 or L2 chains
- Different ZKP based applications, aggregator, AI, oracle



<https://polyhedra.network/>

# *Challenge*



**Experiment with a  
ZKBridge protocol**

# Links, Resources, Literature



ZK Bridges: How Zero-Knowledge Proofs Enable a Cross-Chain World

<https://www.binance.com/en/square/post/889904>

Exploring ZK Bridges

<https://zkv.xyz/exploring-zk-bridges/>

SuperEx Educational Series: Understanding the ZK Bridge Mechanism

<https://superex.medium.com/superex-educational-series-understanding-the-zk-bridge-mechanism-d555891bfd2c>

Zero-Knowledge Proof Bridge

<https://chainscorelabs.com/glossary/smart-contracts/cross-chain-smart-contracts/zero-knowledge-proof-bridge>

zkBridge: Trustless Cross-chain Bridges Made Practical

<https://arxiv.org/pdf/2210.00264>



# *Happy Hunting for the SNARK :)*

## **Q & A**

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